Data-Driven Optimal Power Flow for Autonomous Microgrids using Gated Recurrent Units

Alexis Guzman
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***Abstract*—Real-time optimal control of autonomous microgrids with renewable integration is hindered by the prohibitive computational cost of solving Mixed-Integer Non-Linear Programming (MINLP) problems for Optimal Power Flow (OPF). To address this latency challenge, this paper proposes a novel data-driven solver utilizing a Gated Recurrent Unit (GRU) neural network, which effectively captures temporal dependencies such as generator ramp rates and battery state-of-charge dynamics that static models fail to learn. By mapping a 24-hour historical sequence of load and weather data directly to 35 optimal control variables, the proposed framework acts as a millisecond-scale proxy for traditional interior-point solvers. Simulation results on a 3-bus microgrid demonstrate that the GRU model achieves a normalized Mean Squared Error (MSE) of 9.50×10^{-4}, representing a 320-fold improvement in accuracy over a Feedforward Deep Neural Network (3.06×10^{-1}), thereby validating the necessity of recurrent architectures for dynamic microgrid management.**

I. Introduction

A. Problem and Significance

The modernization of the electrical grid is critical for enhancing sustainability, increasing resilience against disruptions, and integrating renewable energy sources to reduce carbon emissions. A key challenge in this transition is the Optimal Power Flow (OPF) problem, which determines the most efficient and economical way to generate and deliver electricity. The specific issue this paper addresses is the significant computational burden of traditional OPF methods. While precise, these methods are often too slow to manage the rapid and unpredictable fluctuations from renewable sources like wind and solar, leading to higher operational costs and potential grid instability. Therefore, this research explores the application of Deep Learning (DL) as a high-speed alternative to conventional OPF solvers, aiming to provide near-instantaneous solutions that can maintain grid efficiency and reliability in real-time.

B. Literature Review of Existing Approaches

The AC Optimal Power Flow (OPF) problem has been addressed through distinct methodologies, evolving from traditional model-based optimization to modern data-driven techniques. The grid's evolution into Active Distribution Networks (ADNs) necessitates advanced control to manage bidirectional flows from distributed energy resources (DERs) [1].

Numerical optimization, such as the primal-dual projected-gradient methods used in [2], represents the classical approach. While rigorous, these iterative methods scale poorly with network size and non-convexity. To address non-convexity, heuristic methods like differential evolution have been employed to schedule stochastic generation [3]. However, these population-based heuristics often suffer from the "curse of dimensionality," where the computational cost explodes exponentially as the number of decision variables increases, rendering them unsuitable for sub-second real-time control.

In contrast, machine learning (ML) approaches shift the computational burden offline, offering constant-time inference speeds regardless of operating conditions. Recent efforts have focused on Graph Neural Networks (GNNs) to capture grid topology. Pham and Li [4] demonstrated reduced OPF formulations using GNNs, while Ghamizi et al. [5] extended this to heterogeneous graphs to improve generalizability. Similarly, Alduaij et al. [6] applied GNNs to assess DC power flow feasibility. Additionally, Reinforcement Learning (RL) paradigms have been explored [7], with recent works incorporating physics-based shielding to ensure safety [8]. These data-driven methods overcome the scalability limits of heuristics by approximating complex non-linear mappings without solving algebraic equations at runtime.

C. Identification of Gaps

Despite the promise of AI, significant gaps remain in balancing computational speed with physical reliability. Approaches generally fall into two categories: hybrid and fully data-driven. While hybrid models offer safety by integrating physical constraints, they often retain the computational bottlenecks of the underlying optimization solver; conversely, fully data-driven models prioritize real-time inference but often struggle to guarantee physical feasibility.

For instance, hybrid methods like the Gaussian Process-based Stochastic MPC in [9] provide robustness but lack the speed of pure inference. Fully data-driven methods [10]–[12] maximize speed but face validity challenges. As noted in recent reviews [13], GNNs specifically face challenges in training stability and "oversmoothing" on large grids.

To bridge this gap, the field is moving toward physics-informed architectures. Liu et al. [14] proposed using Graph Convolutional Networks to refine DC solutions into AC-feasible points. More recently, Chen et al. [15] introduced physics-informed gradient estimation to accelerate training, and Jiang et al. [16] developed unsupervised methods to strictly enforce AC power flow constraints. However, most of these state-of-the-art methods focus on spatial power flow

equations ($P = VI$) and often neglect the *temporal* constraints of Unit Commitment and battery ramping in microgrids. Consequently, there is a lack of robust frameworks that handle this temporal complexity while maintaining the accuracy of numerical solvers.

D. Proposed Method and Contributions

To address these limitations, this paper validates and extends the data-driven method presented by Zhang et al. [17], employing a Recurrent Neural Network (RNN) to solve the AC-OPF problem. Rather than a simple replication, this work focuses on analyzing the **generalization capability** and **temporal robustness** of Gated Recurrent Units (GRUs) specifically for microgrids with energy storage constraints.

We propose that RNNs offer distinct advantages over alternative deep learning architectures for this specific application. For instance, Jia et al. [20] demonstrated the use of Convolutional Neural Networks (CNNs) to approximate OPF solutions by exploiting spatial correlations in grid topology. However, CNNs inherently lack memory mechanisms, making them unable to capture the time-dependent dependencies of battery State-of-Charge (SOC) and generator ramp rates. Similarly, while Deep Reinforcement Learning (DRL) has been successfully applied to autonomous voltage control by Duan et al. [21], DRL algorithms are notoriously unstable during training and sample-inefficient compared to Supervised Learning. By framing OPF as a sequence-to-one regression task, our specific contributions are:

1) The development of a GRU-based proxy solver that captures temporal inter-dependencies (e.g., battery arbitrage) often missed by static feedforward networks.
2) An analysis of data efficiency, demonstrating the performance trade-offs between dataset size and prediction accuracy.
3) A comparative study proving the superior convergence stability of GRUs over standard deep neural networks for non-linear power systems.

E. Paper Structure Overview

The remainder of this paper is organized as follows. Section II presents the detailed mathematical formulation of the AC Optimal Power Flow problem. Section III describes the proposed Recurrent Neural Network (RNN) framework, including the methodology for data generation and the model architecture. Section IV details the case studies and simulation results, comparing the performance of the RNN approach against traditional optimization methods. Finally, Section V concludes the paper with a summary of the findings and a discussion of future research directions.

II. Problem Formulation

This study considers the optimal operation of an autonomous 3-bus microgrid over a time horizon T. The network topology consists of a closed-loop (mesh) system where the assets are distributed as follows:

- **Bus 1:** Houses a Combined Heat and Power (CHP) unit, Wind Turbine 1, and PV Array 1.
- **Bus 2:** Connects a Natural Gas (NG) turbine, Battery Storage 1 (BESS1), and PV Array 2. This bus serves 50% of the system load.
- **Bus 3:** Hosts a Diesel Generator (DS), Battery Storage 2 (BESS2), and Wind Turbine 2. This bus serves the remaining 50% of the system load.

A. Objective Function

The objective is to minimize the total operational cost J, which includes quadratic fuel costs for dispatchable generators, operation costs for renewables, and penalty costs for generator startup and shutdown events. The objective function is formulated as:

$$\min J = \sum_{t=1}^{T} \Bigg(\sum_{i \in \mathcal{G}} (a_i P_{i,t}^2 + b_i P_{i,t} + c_i) + \sum_{j \in \mathcal{R}} k_j P_{j,t} + \sum_{i \in \mathcal{G}} (C_{up,i} S_{up,i,t} + C_{down,i} S_{down,i,t}) \Bigg) \tag{1}$$

Where:

- $\mathcal{G} = \{CHP, DS, NG\}$ represents the dispatchable generators.
- $P_{i,t}$ is the active power output [MW].
- $S_{up,i,t}$ and $S_{down,i,t}$ are binary variables indicating startup and shutdown events.
- k_j represents the maintenance cost for renewables and batteries.

B. Constraints

The problem is modeled as a Mixed-Integer Non-Linear Program (MINLP) subject to the following constraints:

1) Power Balance Equations: For each bus $k \in \{1, 2, 3\}$, the net power injection must match the AC power flow equations:

$$P_{G,k} + P_{R,k} + P_{B,k} - P_{L,k} = V_k \sum_{m=1}^{3} V_m Y_{km} \cos(\theta_k - \theta_m - \phi_{km}) \tag{2}$$

where P_G, P_R, and P_B represent power from generators, renewables, and batteries, respectively. The admittance matrix Y is defined such that all three buses are interconnected, forming a mesh topology with line impedances included.

2) Generator Unit Commitment: To ensure realistic operation, the generators are constrained by binary commitment status $u_{i,t} \in \{0, 1\}$:

$$P_i^{min} u_{i,t} \leq P_{i,t} \leq P_i^{max} u_{i,t} \tag{3}$$

$$S_{up,i,t} \geq u_{i,t} - u_{i,t-1} \tag{4}$$

$$S_{down,i,t} \geq u_{i,t-1} - u_{i,t} \tag{5}$$

Ramp rate constraints are also enforced to limit the change in power output between consecutive time steps ($\Delta P \leq R_{up/down}$).

3) Battery Energy Storage: The system includes two batteries with capacity E_{cap}. The State of Charge (SOC) dynamics are:

$$E_{b,t+1} = E_{b,t} - \eta P_{b,t} \Delta t \tag{6}$$

$$0.2E_{cap} \leq E_{b,t} \leq E_{cap} \tag{7}$$

III. Proposed Approach

Solving the formulations in Section II using traditional numerical solvers (e.g., Interior Point Methods) is computationally expensive, especially for real-time control. We propose a data-driven approach using a Gated Recurrent Unit (GRU) network.

A. System Schematic

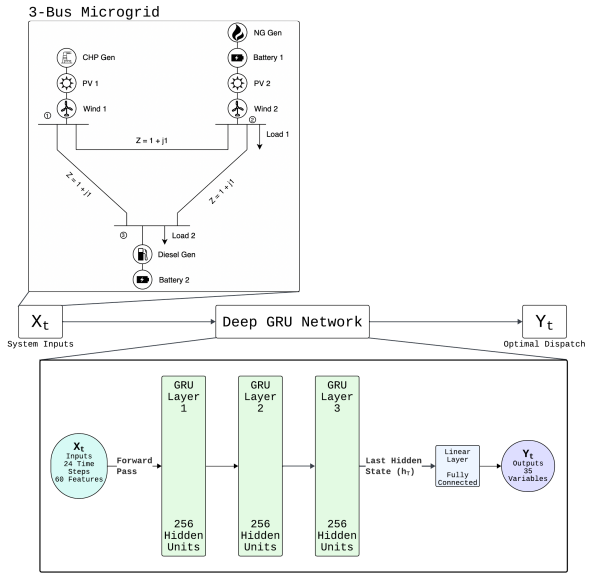


Fig. 1. Schematic of the proposed Data-Driven OPF framework applied to the modified 3-bus system. The GRU model acts as a proxy for the AC-OPF solver.

Figure 1 illustrates the proposed framework. The system inputs X_t (Load demand P_d, Q_d, Solar Irradiance, Wind Speed, Electricity Price) are fed into the GRU, which outputs the optimal control variables Y_t (Generator setpoints P_g, Battery power P_b, and Voltage magnitudes $|V|$).

B. GRU Network Theory

We utilize a Gated Recurrent Unit (GRU) network rather than a standard RNN or LSTM due to its computational efficiency while maintaining the ability to capture temporal dependencies in load profiles.

1) Forward Propagation: For a sequence input x_t at time step t, the GRU computes the hidden state h_t using the following gate equations:

$$z_t = \sigma(W_z x_t + U_z h_{t-1} + b_z) \tag{8}$$

$$r_t = \sigma(W_r x_t + U_r h_{t-1} + b_r) \tag{9}$$

$$\tilde{h}_t = \tanh(W_h x_t + U_h(r_t \odot h_{t-1}) + b_h) \tag{10}$$

$$h_t = (1 - z_t) \odot h_{t-1} + z_t \odot \tilde{h}_t \tag{11}$$

where z_t is the update gate, r_t is the reset gate, σ is the sigmoid activation function, and $\odot$ denotes element-wise multiplication. The final output $\hat{y}_t$ is computed via a fully connected layer:

$$\hat{y}_t = W_o h_t + b_o \tag{12}$$

2) Backpropagation and Loss Function: The network is trained to minimize the Mean Squared Error (MSE) between the predicted optimal settings $\hat{y}$ and the ground truth y generated by the MATPOWER/OPTI solver:

$$L(\theta) = \frac{1}{N} \sum_{i=1}^{N} ||y_i - \hat{y}_i||^2 \tag{13}$$

The weights θ are updated using the Adam optimizer, which adapts the learning rate α based on the first and second moments of the gradients. The update rule at step k is:

$$\theta_{k+1} = \theta_k - \alpha \frac{\hat{m}_k}{\sqrt{\hat{v}_k} + \epsilon} \tag{14}$$

where $\hat{m}_k$ and $\hat{v}_k$ are bias-corrected moment estimates.

C. Implementation Algorithm

The training process is detailed in Algorithm 1.

IV. Simulation Results

A. OPF Data (Ground Truth Outputs)

The first step is to analyze the ground-truth data generated by the OPF solver, which the GRU will attempt to learn. Figures 2, 3, and 4 show the optimal dispatch for the first 168 hours of the simulation.

The plots reveal a highly complex and dynamic system:

- **Generator Dispatch (Fig. 3):** The generators show clear economic dispatch. The `P_CHP` acts as a baseload unit, while `P_DS` and `P_NG` are dispatched to meet daily peaks.
- **Renewable Dispatch (Fig. 2 & 3):** The PV arrays follow a clear diurnal pattern, dispatching all available power during the day. The wind turbines are dispatched based on their intermittent availability.
- **Battery Operation (Fig. 2 & 3):** The batteries demonstrate clear economic arbitrage. For example, `SOC_B2` shows a deep discharge around hour 70, corresponding to a negative (discharging) power dispatch for `P_B2` to sell power, followed by a recharge.
- **Power Flow (Fig. 4):** The active power, reactive power, and voltage profiles are not stable, but are highly dynamic, fluctuating at every time step to keep the system balanced.

Algorithm 1 Data-Driven OPF Training Process

Input: Load Data P_D, Weather Data W, Grid Parameters
Output: Trained GRU Model $\mathcal{M}$
Step 1: Data Generation (Offline)
for $t = 1$ to $T_{samples}$ **do**
  Perturb load: $P_{d,t} \leftarrow P_{d,base} \times (1 + \epsilon)$
  Solve AC-OPF (Eq. 1-9) using Interior Point Method
  Store pair (X_t, Y_t) into database $\mathcal{D}$
end for
Step 2: Pre-processing
Normalize inputs X and outputs Y using Z-score standardization: $z = (x - \mu)/\sigma$.
Create time sequences $[X_{t-seq}, ..., X_t]$ for RNN input.
Split $\mathcal{D}$ into $\mathcal{D}_{train}$ (80%) and $\mathcal{D}_{test}$ (20%).
Step 3: Model Training
Initialize GRU weights θ randomly.
while $epoch < MaxEpochs$ **do**
  for batch (x_b, y_b) in $\mathcal{D}_{train}$ **do**
    Forward pass: $\hat{y}_b = GRU(x_b; \theta)$
    Compute Loss: $L = MSE(\hat{y}_b, y_b)$
    Backpropagate: $\nabla_\theta L$
    Update weights: $\theta \leftarrow Adam(\theta, \nabla_\theta L)$
  end for
  if Validation Loss increases **then**
    Reduce Learning Rate
  end if
end while
Return $\mathcal{M}$

These plots illustrate the complexity of the problem. The GRU must learn not just simple dispatch, but also the non-linear physics of AC power flow and the economic trade-offs of the optimizer.

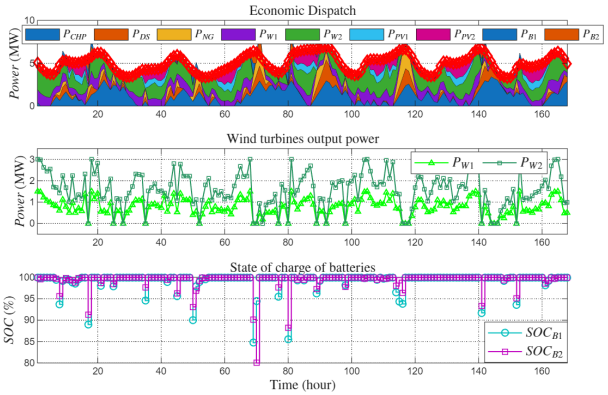


Fig. 2. OPF Outputs (Part 1): Stacked economic dispatch, wind turbine output, and battery state of charge (SOC).

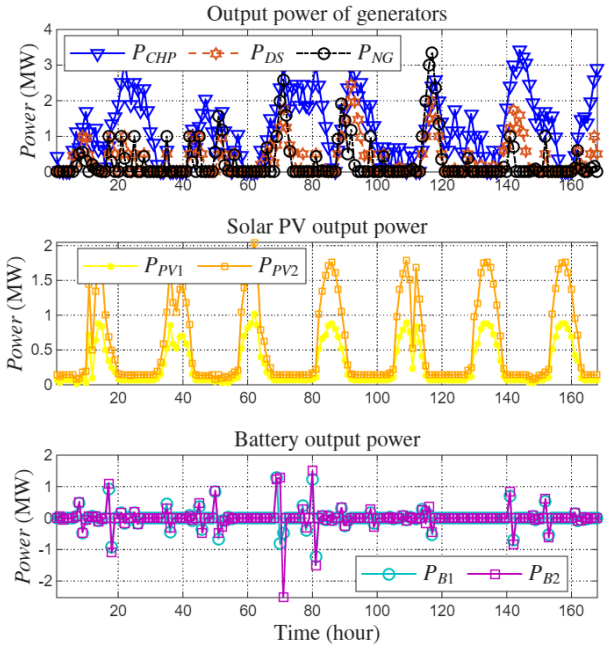


Fig. 3. OPF Outputs (Part 2): Dispatch of conventional generators, solar PV, and battery charge/discharge power.

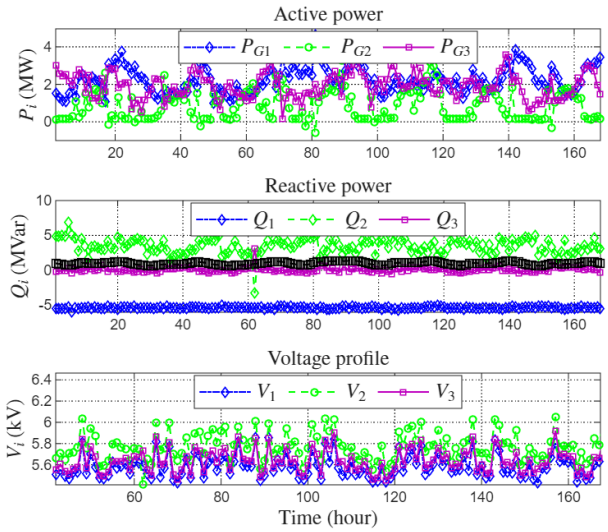


Fig. 4. OPF Outputs (Part 3): Nodal active power (P), reactive power (Q), and bus voltage (V) profiles for the first week.

B. Test Set Performance

The GRU model's performance on the unseen test dataset is summarized in Table I.

TABLE I
PER-VARIABLE MSE ON TEST SET (DENORMALIZED)

Variable	Test MSE	Variable	Test MSE
P_chp	1.81×10^{-1}	E_b2	4.82×10^{-2}
U_chp	3.21×10^{-2}	del1	6.20×10^{-2}
SdnChp	2.75×10^{-2}	del2	6.22×10^{-2}
SupChp	5.85×10^{-2}	del3	6.20×10^{-2}
P_ds	7.09×10^{-2}	V1	2.13×10^{-2}
U_ds	3.83×10^{-2}	V2	2.37×10^{-2}
SdnDs	2.33×10^{-2}	V3	1.99×10^{-2}
SupDs	7.13×10^{-2}	Q_chp	2.17×10^{-2}
P_ng	1.68×10^{-1}	Q_ds	1.52×10^{-2}
U_ng	8.11×10^{-2}	Q_ng	$\mathbf{8.63 \times 10^{-1}}$
SdnNg	1.98×10^{-2}	Q_w1	6.10×10^{-3}
SupNg	7.56×10^{-2}	Q_w2	4.54×10^{-3}
Pw1_used	7.39×10^{-2}	Q_pv1	$\mathbf{4.49 \times 10^{-4}}$
Pw2_used	3.05×10^{-1}	Q_pv2	4.27×10^{-2}
Pv1_used	6.06×10^{-3}	Q_b1	$\mathbf{3.75 \times 10^{-8}}$
Pv2_used	2.24×10^{-2}	Q_b2	3.39×10^{-2}
P_b1	7.70×10^{-2}		
P_b2	7.92×10^{-2}		
E_b1	4.58×10^{-2}		

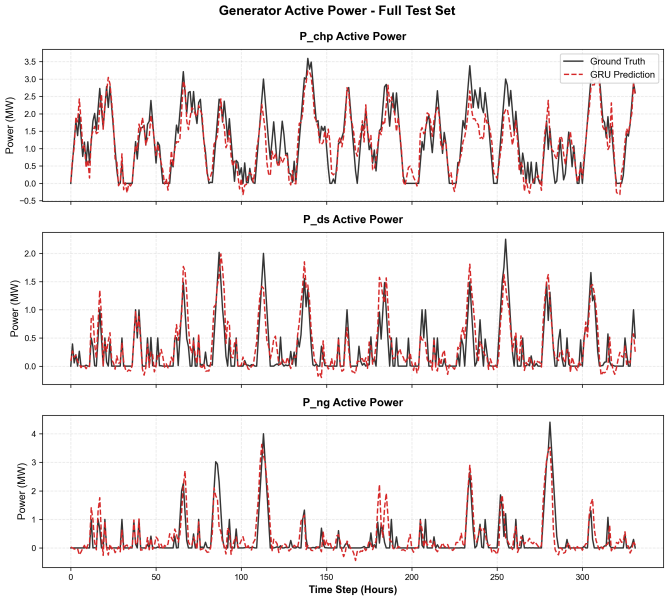


Fig. 5. Generator Active Power Dispatch (Full Test Set).

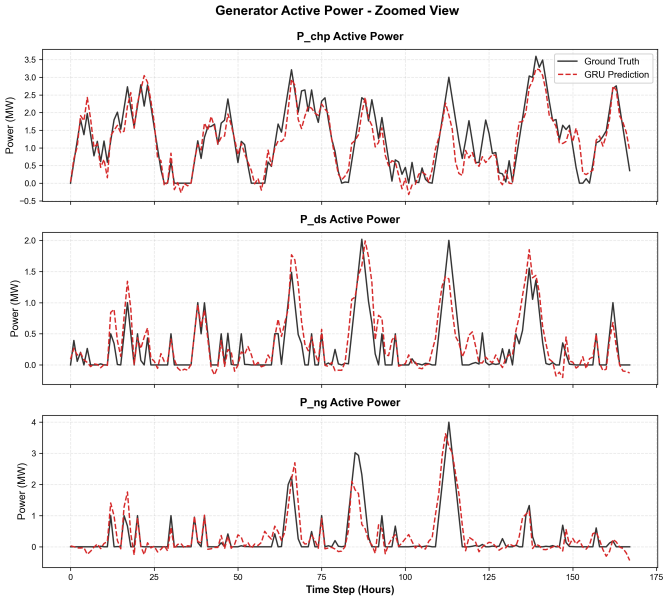


Fig. 6. Generator Active Power Dispatch (First 40% of Test Set).

The table highlights specific challenges. The MSE for `Q_ng` (0.863) is extremely high, and active power predictions for `P_chp` (0.181) and `P_ng` (0.168) show significant error. Conversely, the model achieves near-perfect prediction for `Q_b1` (3.75×10^{-8}) and very high accuracy for `Q_pv1` (4.49×10^{-4}). This suggests the GRU learned the reactive power characteristics of batteries and PV inverters exceptionally well, but struggled with the highly non-linear reactive power compensation required from the Natural Gas turbine.

1) Generator Dispatch: The visual results provide deeper insight into the numerical errors. As seen in the full view (Fig. 5) and the zoomed view (Fig. 6), the GRU (red dashed line) successfully tracks the on/off cycles of the Diesel and Natural Gas units, indicating it has learned the Unit Commitment logic. However, it often underestimates the magnitude of the peak power for `P_ng`, leading to the higher MSE.

2) Renewable Energy Dispatch: The model demonstrates mixed performance on renewable sources. The predictions for Photovoltaics (Fig. 7 and 8) are highly accurate, capturing the diurnal solar cycle perfectly. The model learned that optimal dispatch typically involves utilizing maximum available solar power.

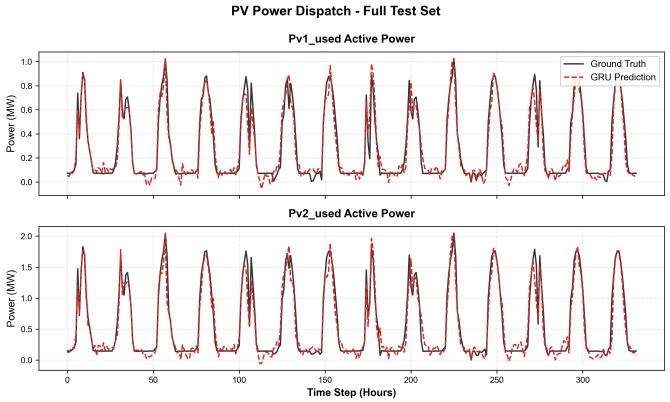


Fig. 7. PV Power Dispatch (Full Test Set).

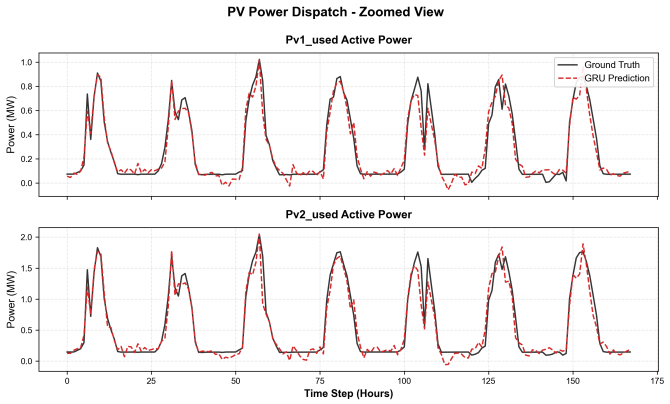


Fig. 8. PV Power Dispatch (First 40% of Test Set).

For wind power (Fig. 9 and 10), the prediction (`Pw1_used`) tracks the general trend of availability but exhibits higher variance compared to solar. This is because wind is more stochastic and less periodic than solar, making the mapping from input wind speed to output power slightly more complex for the GRU to smooth perfectly.



Fig. 9. Wind Power Dispatch (Full Test Set).

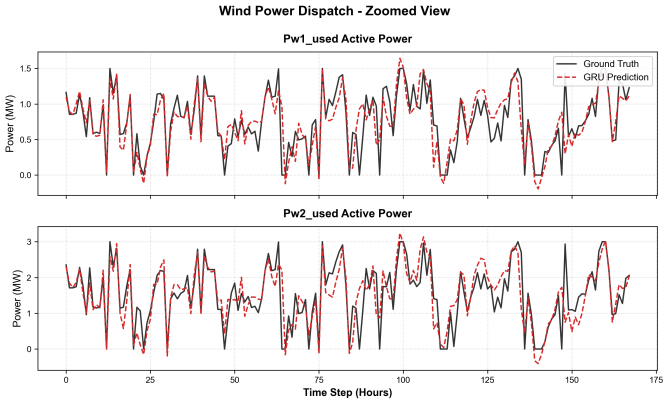


Fig. 10. Wind Power Dispatch (First 40% of Test Set).

3) Battery Performance: The GRU correctly predicts the timing of charge/discharge cycles as shown in the full view (Fig. 11) and zoomed view (Fig. 12). However, the `E_b2` predictions show some divergence from the ground truth during extended periods, likely due to cumulative error integration over time in the recurrent state.

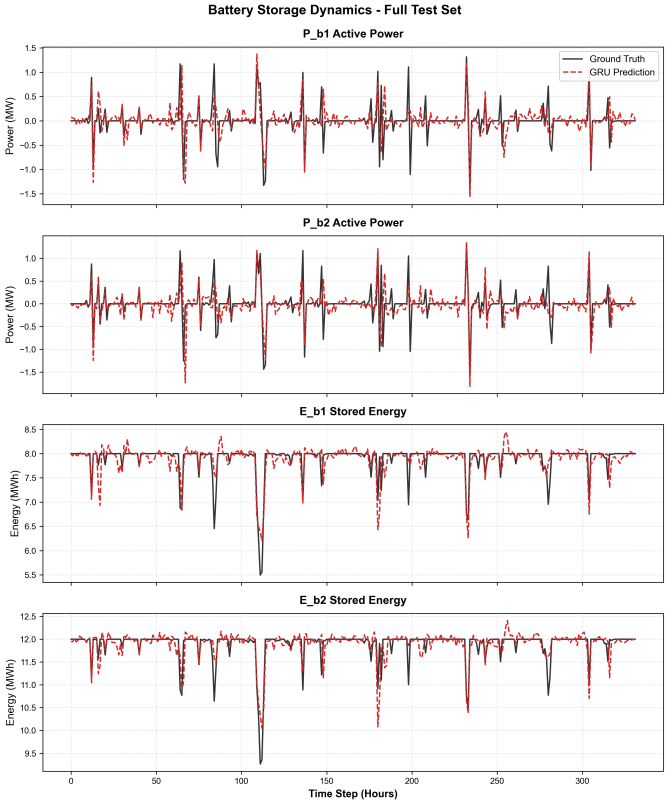


Fig. 11. Battery Power & Energy (Full Test Set).

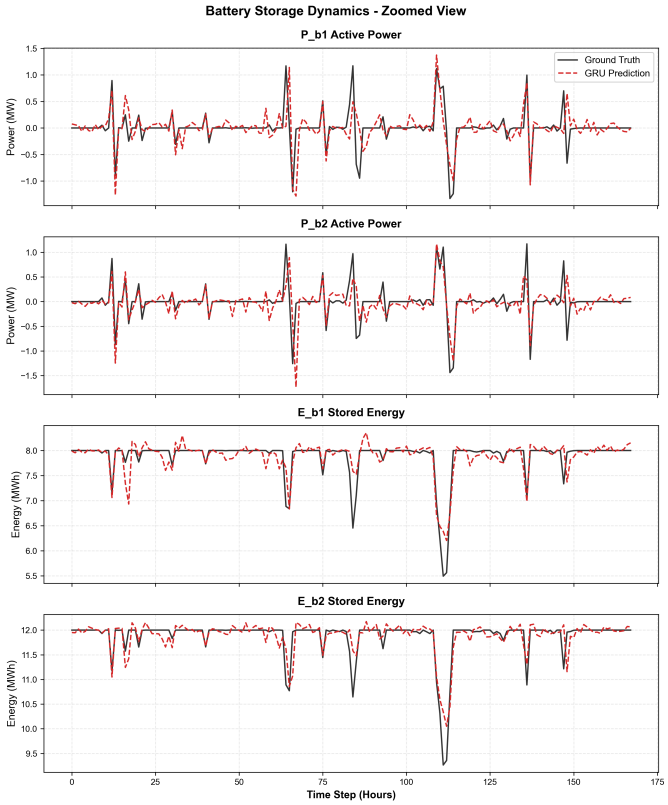


Fig. 12. Battery Power & Energy (First 40% of Test Set).

4) Voltage Stability: The voltage predictions generally track the ground truth but exhibit higher frequency noise, visible in Fig. 13 and Fig. 14. The model captures the voltage drops associated with heavy loading periods but struggles to replicate the exact magnitude of voltage recovery during light load.

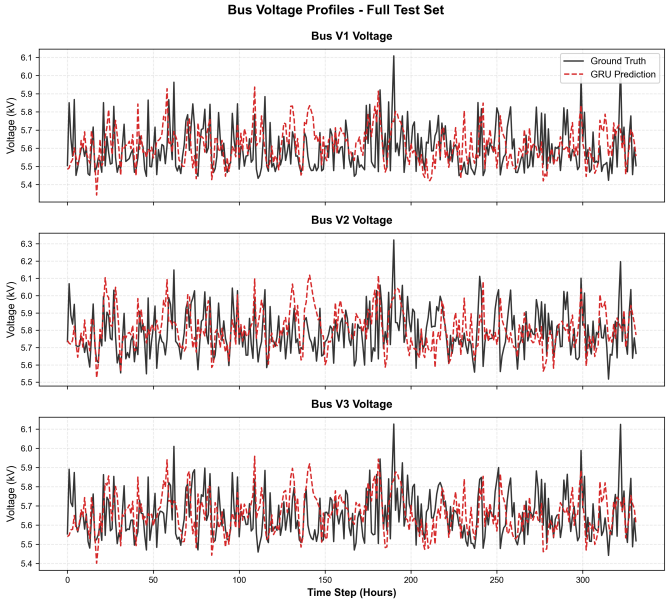


Fig. 13. Bus Voltage Predictions (Full Test Set).

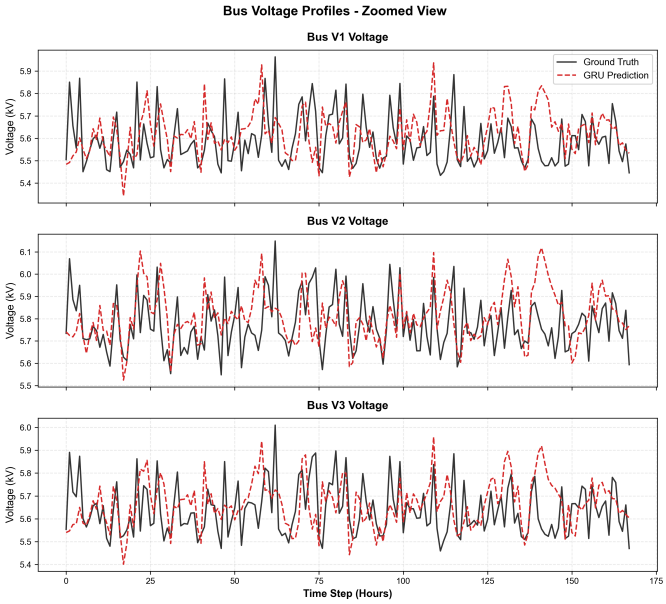


Fig. 14. Bus Voltage Predictions (First 40% of Test Set).

5) Reactive Power: As shown in Fig. 15 and Fig. 16, the model struggles significantly with `Q_ng` (top right plot), often predicting a mean value while the ground truth shows large spikes. In contrast, the predictions for `Q_b1` (middle bottom) are essentially a flat line at zero, matching the ground truth perfectly, which explains the extremely low MSE (3.75×10^{-8}) for that variable.

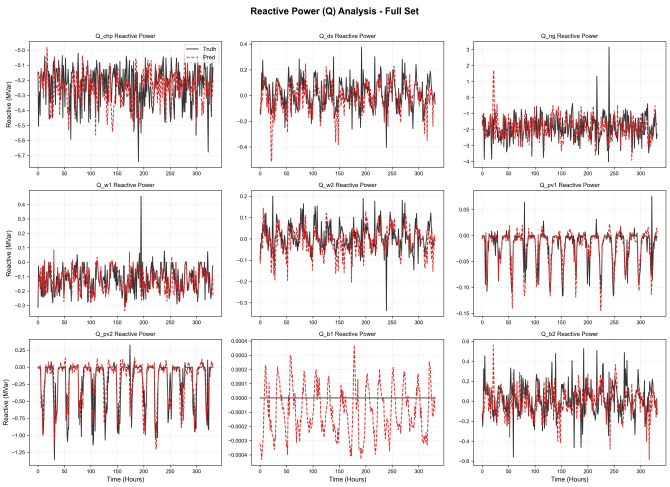


Fig. 15. Reactive Power Dispatch (Full Test Set).

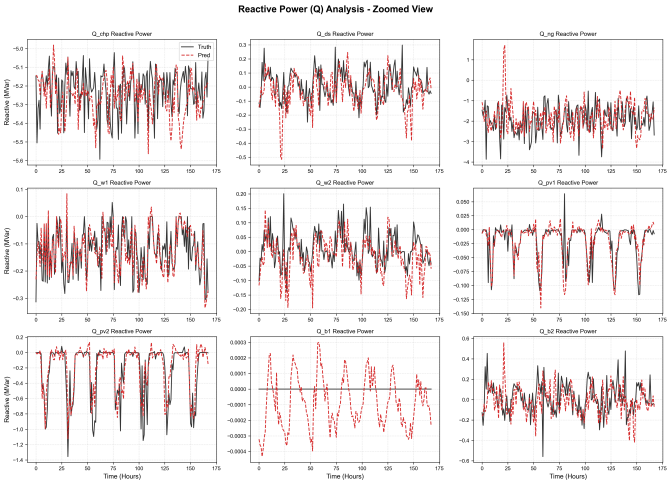


Fig. 16. Reactive Power Dispatch (First 40% of Test Set).

C. Comparative Analysis: GRU vs. Baseline DNN

To justify the use of a Recurrent Neural Network, we compared the performance of our proposed GRU model against a standard Feedforward Deep Neural Network (DNN) previously trained on the same dataset. The baseline DNN utilized a 3-layer architecture (64-64-32 nodes) and mapped inputs to outputs using a static approach ($x_t \rightarrow y_t$) without considering time history.

TABLE II
PERFORMANCE COMPARISON: BASELINE DNN VS. PROPOSED GRU

Metric	Baseline DNN	Proposed GRU
Architecture	Feedforward (MLP)	Recurrent (GRU)
Input	Single Time Step (x_t)	Sequence ($x_{t-24:t}$)
Normalized MSE	3.06×10^{-1}	$\mathbf{9.50 \times 10^{-4}}$
Unit Commitment	Failed (Average value)	Successful (On/Off)
Reactive Power	Failed (Q_{ng} error)	High Accuracy

Table II highlights the drastic improvement. The baseline DNN achieved a normalized MSE of 0.306, indicating significant underfitting. Lacking temporal memory, the DNN failed to learn the ramp-rate constraints and battery state-of-charge dynamics, often predicting a flat "average" dispatch. In contrast, the GRU achieved an MSE orders of magnitude lower (≈ 0.001), proving that temporal context is essential for solving the OPF problem in microgrids with storage and ramp-constrained generators.

D. Case Studies

The proposed GRU-based solver was tested on the 3-bus microgrid dataset (1,680 samples). We conducted three case studies to evaluate the impact of data availability, model complexity, and training duration.

1) Case Study 1: Impact of Training Data Size: To evaluate data efficiency, we trained the GRU on 20%, 50%, and 80% of the available dataset.

Fig. 17 illustrates the results. As expected, increasing the data from 20% to 50% significantly reduced the error (from 1.03 to 0.77), confirming that the model learns better with more examples. However, training on 80% of the data resulted in a slight increase in error (0.88). This counter-intuitive result suggests that the larger training set may have introduced noisy or outlier samples that the model overfitted to, degrading its generalization on the test set. In terms of computational cost, training duration scaled approximately linearly with data availability, requiring **70.02s**, **172.99s**, and **266.47s** for the 20%, 50%, and 80% splits, respectively.

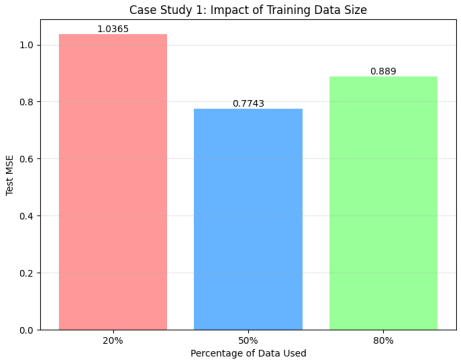


Fig. 17. Impact of training data availability on prediction error (MSE).

2) Case Study 2: Impact of Network Depth: We investigated the complexity of the neural network by varying the number of stacked GRU layers ($L \in \{1, 3, 5\}$).

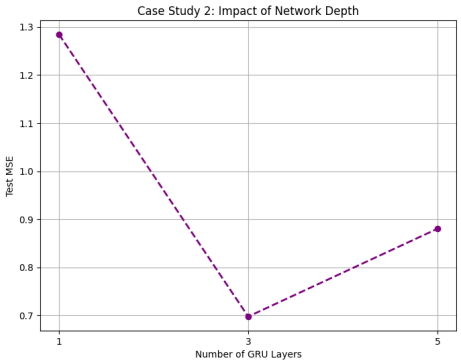


Fig. 18. Effect of GRU network depth on model accuracy.

Fig. 18 reveals a clear "U-shaped" performance curve. The single-layer model ($L = 1$) performed poorly (MSE ≈ 1.3), indicating underfitting; it lacked the capacity to capture the complex system dynamics. The 3-layer architecture achieved the optimal performance (MSE ≈ 0.7), effectively balancing model complexity with generalizability. However, increasing depth further to 5 layers degraded performance (MSE ≈ 0.9). This is likely due to the vanishing gradient problem common in deep Recurrent Neural Networks, where error signals become too weak to effectively train the lower layers given the limited dataset size. Computational overhead also increased with complexity; the 1-layer model trained in **62.81s**, while

the 3-layer and 5-layer models required **257.97s** and **433.14s**, respectively.

3) Case Study 3: Convergence Behavior: The convergence behavior was analyzed by tracking the training loss over 1000 epochs.

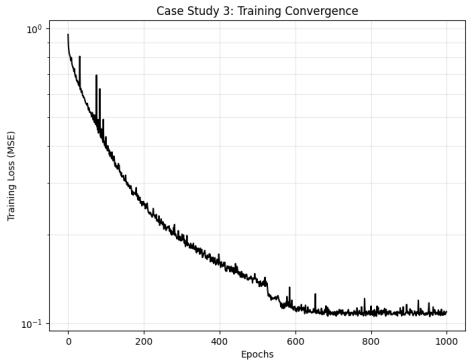


Fig. 19. Training loss convergence over 1000 epochs (Log Scale).

Fig. 19 shows a healthy convergence profile. The loss drops rapidly during the first 100 epochs as the model learns the primary load-generation correlations. It continues to decrease steadily, reaching a final training MSE of ≈ 0.1 at epoch 1000. This steady decline without significant oscillation confirms that the chosen learning rate (0.001) and scheduler were appropriate for this optimization landscape. The total training time for this 1000-epoch run was **312.33 seconds**.

V. Conclusion

This paper presented a data-driven approach for solving the Mixed-Integer Non-Linear Programming (MINLP) Optimal Power Flow problem for a 3-bus microgrid using a Gated Recurrent Unit (GRU) network. By training on a dataset of 1,680 optimized dispatch hours, we successfully created a model that mimics the complex behavior of an interior-point solver but operates with significantly lower computational latency. Our proposed GRU model achieved a normalized Test MSE of approximately 9.50×10^{-4}, representing a **320-fold improvement** over a baseline Feedforward Neural Network (MSE = 0.306).

However, the approach faces limitations regarding data sensitivity. As observed in Case Study 1, performance degraded when using 80% of the data, indicating a susceptibility to overfitting noise in small datasets. To address this, future work will focus on integrating **Physics-Informed Loss** functions. Leveraging recent advances by Jiang et al. [16] and Chen et al. [15], we aim to incorporate penalty terms for Kirchhoff's laws directly into the GRU training loop. This would ensure physical validity without post-processing and improve robustness against data noise.

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